pfi

# Python for Fixed Income

COURSE 9 · DEPLOYMENT

# The Credit Desk as a Service

WEEK 2 · DAY 3

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This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

# Today: the desk as a service

## Today

1. The desk without a key: Course 8 in replay
2. Why a snapshot: size, and what replay answers
3. desk_snapshot: the brief, every exchange hashed
4. desk_app: four endpoints, the refusal first
5. The live switch, and the platform's secrets
6. The guardrail, checked on the served JSON
7. Deploying it: the course's deploy, recorded
8. The deploy log

## Nothing is sent

**To a model**

the brief replays from saved runs

**To Vercel**

the cloud steps are recorded

**The server never holds a key.**

**Freeze what the desk returned, trace every figure, and serve that.**

# What the deployed desk serves, and never

| | |
|---|---|
| **Descriptive only** | what documents state, the curve between dates, the book's DV01 |
| **Every response dated** | figures as of 2026-10-02, next year too |
| **A view is refused first** | before anything is looked up |
| **No key, synthetic holdings** | a real company only as a document's author |

## What the service serves

### Allowed names

the brief as of 2026-10-02

the check table

the saved run

### Never called

the desk's view

advice or a call

a recommendation

# The desk without a key

```
OpenAI key loaded: False. Mode: replay.
```

| 59 | 10 | 21 | 53 | 2 |
|---|---|---|---|---|
| saved exchanges | covenant rows | brief requests | checks passed | checks failed |

### The run line: where every figure came from

Model gpt-4.1-mini-2025-04-14, temperature 0, seed 42; 21 requests, 39,691 prompt and 6,154 completion tokens;

saved run recorded 2026-10-11; replayed (no key loaded).

**The two failures are Course 8's own. A service serves them beside the passes.**

# Why a snapshot

| | Replay service | Snapshot service |
|---|---|---|
| Requirements | 20 | 1 (fastapi) |
| Distributions | 94 | 11 |
| Installed here (Windows) | 923.4 MB | 9.6 MB |
| Linux, read by day 4 | about 5,716.5 MB | about 9.2 MB |

Largest in the replay: torch 452.8 MB, scipy 102.8 MB, transformers 47.9 MB

### A question nobody recorded, to the replay

```
LookupError: no saved exchange for request ab75aca0dc68 in day15
```

**Vercel's bundle limit is 500 MB. And a replay answers only what was recorded: 3 graded answers.**

# desk_snapshot: the brief, frozen as data

### desk_snapshot.json

25,853 bytes of JSON

as_of 2026-10-02, mode replay

the brief and 55 checks

the run line, the disclaimer

3 graded answers

21 keys, each with a SHA-256

Course 8's input rule, as data

### Every figure traces to a file

a figure in the brief

-> names the tool request that computed it

-> which lives in a saved exchange

-> a file whose SHA-256 anyone can recompute

**checked in the notebook: 21 of 21**

**An exchange missing from the folder raises ValueError: a snapshot holds only what was saved.**

# desk_app: four endpoints, the refusal first

| Endpoint | Returns |
|---|---|
| GET /health | version, as_of, mode |
| GET /brief | the three sections, as_of |
| GET /checks | the 55 checks, as_of |
| POST /ask | see the order on the right |

### POST /ask, in order

1. a view? 200, refused, the refusal line

2. recorded? 200, its saved answer

3. anything else: 404, not in the saved runs

```
200 refused: True rule: opinion words <- Which of these bonds should the desk buy?
200 refused: True rule: an ordering of securities <- Rank the three issuers' indentures.
200 refused: True rule: opinion words <- Where will the 10-year yield be next month?
200 refused: True rule: a real issuer beside a holding <- Is Lamar a holding in this portfolio?
```

**Status 200 for a refusal: the request was understood and answered. A 4xx would call it malformed.**

# The live switch, and where secrets live

```
nothing set, as on the course's deployments  {'mode': 'replay', 'key_loaded': False}
DESK_MODE live, no key  {'mode': 'replay', 'key_loaded': False}
DESK_MODE live and a key  {'mode': 'live', 'key_loaded': True}
```

The key in any response: no. The settings hold key_loaded, never the key.

On a platform: environment variables, in the project's settings, never in a file.

Vercel: a changed environment variable applies only to new deployments (vercel.com/docs/environment-variables, read 2026-10-12).

Vercel: a Secret is write-only once saved (vercel.com/docs/environment-variables/sensitive-environment-variables, read 2026-10-12).

**The course sets neither DESK_MODE nor a key. Going live is your own key, and your own budget.**

# The guardrail at the API

| Served section | Real issuers | Synthetic issuers | CUSIPs | Document ids |
|---|---|---|---|---|
| macro | 0 | 0 | 0 | 0 |
| documents | 0 | 0 | 0 | 2 |
| portfolio | 0 | 0 | 0 | 0 |

Course 8's separation checks, as served: 4 of 4 passed.

**Nothing served puts a real company beside a holding: checked on the JSON a stranger receives.**

# Deploying it

### pfi-desk-service

```
app.py
bondmath.py
deploykit.py
desk_snapshot.json
requirements.txt (fastapi)
.python-version (3.13)
README.md
```

### What the platform printed, masked

```
Recorded 2026-10-12, on the course's account, masked
Directory ~\projects\pfi-desk-service
Detected FastAPI (Output Directory: N/A)
? Created student/pfi-desk-service
Previous build caches not available.
Build Completed in /vercel/output [8s]
? Aliased https://student.vercel.app
? Ready in 21s
```

Vercel Python: project files reachable at build time are bundled (vercel.com/docs/functions/runtimes/python, read 2026-10-12).

# The deployed service: /brief and /ask

```
> curl.exe -s -w "`nHTTP status %{http_code}`n" https://student.vercel.app/brief
{"as_of":"2026-10-02","mode":"replay","brief":{"macro":{"text":"The FOMC post-meeting statement o...
HTTP status 200
the deployed /brief is the test client's /brief here, every figure: yes
```

```
> $body = '{"question": "Which of these bonds should the desk buy?"}'; $r = Invoke-WebRequest -Us...
HTTP status 200
X-Desk-Refused: true
{"question":"Which of these bonds should the desk buy?","answer":"REFUSED: this course asks only ...
```

**The snapshot travelled with the app; the rule ran on Vercel with no agentkit installed.**

# Excel side by side: what the recording cost

**Excel side by side (tested in Excel)**

| | |
|---|---|
| `B2:B3, tokens of 59 exchanges` | 110,907 and 11,833 |
| `C2:C3, prices per token` | 0.0000004 and 0.0000016 |
| `=SUMPRODUCT(B2:B3,C2:C3)` | 0.0632956000 |
| `=B2*C2+B3*C3` | 0.0632956000 |
| `=SUMPRODUCT(B2:B3,D2:D3)` | 0 (D2:D3: prices as text) |

**textkit.usage_cost: 0.0632956000, equal within 1e-12. Type prices as numbers.**

# The AI check

The docstring is the prompt. The deploy code that comes back runs and looks finished.

1

## Docstring

Units, conventions, and what the code must never let through: what you asked for.

→

2

## Code back

Written for the course in the style of an AI answer. Read it.

→

3

## Test first

Plant the case the docstring promises to catch. Run it: it fails.

→

4

## Fix

Find the rule it broke, fix it, run the test again.

**Today: the settings keep the key "for debugging", and /health returns them to anyone.**

*Code that runs is not the same as code that is right.*

# What you can do now

- Replay the desk with no key; a replay answers only what it recorded
- Ship a snapshot: one requirement, every exchange hashed
- desk_app: dated responses, 404s with a reason, refusals first
- live_settings: a mode and a yes or no, never a value
- The guardrail on the served JSON; 33 tests; the deploy logged

**Week 2, Day 4** Streamlit and Gradio front ends